Reliability-aware active learning for FFLO phase spaces

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Resolving finite-momentum superconducting phases across expanding parameter spaces requires both reliable free-energy minimization and controlled use of expensive exact calculations. We retain an audited Bogoliubov–de Gennes (BdG) oracle as the sole source of thermodynamic and topological labels, while active learning only schedules the next exact evaluations. This separation yields an audited two-dimensional reconstruction of Normal, uniform-superconducting, and FFLO structure while keeping a hard-risk numerical layer explicit. On trusted gapped FFLO states, full-zone gap and Pfaffian diagnostics resolve connected cFFLO/tFFLO structure that passes the configured $k = 8$ audit with a retained $k = 6$ sensitivity. The same framework exposes thermodynamic and topology-aware structure in $(k_BT/t, J_A/t, \mu/t)$ across a finite sampled domain; the exact support is reported discretely, without inferring a continuous three-dimensional topology surface or a complete atlas.

INTRODUCTION

Superconductivity with nonzero center-of-mass pairing momentum provides a route beyond a spatially uniform condensate. Fulde and Ferrell and, independently, Larkin and Ovchinnikov established the finite-momentum pairing and spatially modulated mechanisms now grouped under the FFLO setting [1, 2]. Subsequent reviews have developed the thermodynamic, material, and ultracold-gas consequences of Fermi-surface mismatch and competing pairing momenta [3–5]. The central phase-selection problem is therefore intrinsically thermodynamic: competing minima must be compared over both pairing amplitude and momentum.

The search for FFLO structure spans several physical platforms. Heavy-fermion and organic superconductors have displayed field-induced phases and spectroscopic, calorimetric, or magnetic signatures discussed in relation to FFLO order [6–10]; these representative studies motivate the problem without implying that every identification is universal or uncontested. Population-imbalanced ultracold Fermi gases provide a complementary tunable setting for pairing under mismatch, phase separation, and finite-momentum phases [11–15].

Altermagnets add a distinct route to this competition. Their compensated magnetic order can nevertheless generate symmetry-structured momentum-dependent spin splitting [16–19]. Model studies have connected this splitting to zero-field finite-momentum or other unconventional pairing states [20–22], and neighboring altermagnetic constructions have been proposed for topological superconductivity and Majorana modes [23, 24]. These precedents motivate an altermagnetic finite-q problem but do not determine the phase structure of the one-dimensional model studied here.

Topology adds a second, logically separate requirement. Finite-momentum pairing can coexist with topological superfluidity in spin-orbit-coupled models [25–27]. More generally, bulk symmetries and invariants organize topological superconducting phases [28–32]. Majorana-platform studies connect those bulk distinctions to boundary physics [33–36]. In a gapped one-dimensional Bogoliubov–de Gennes (BdG) system in symmetry class D, the relevant $\mathbb{Z}_2$ distinction can be expressed through a Pfaffian invariant [37, 38]; its use is meaningful only after bulk-gap applicability is established. A topology-resolved FFLO map must therefore identify a reliable thermodynamic minimum before classifying trusted gapped FFLO states.

This hierarchy is computationally demanding because every accepted parameter point requires minimization of $F(\Delta, q)$, comparison with the normal reference, adaptive checks of shallow or edge minima, and, where applicable, full-zone spectral and Pfaffian diagnostics. Active learning and Bayesian experimental design formalize the selection of informative sequential evaluations [39–41]; phase-diagram applications demonstrate boundary-focused sampling [42–44], while adaptive sampling has also reduced expensive electronic-structure, atomistic, and chemical search budgets [45–50]. For a physical phase diagram, however, an acquisition model should decide where to calculate next, not replace the exact calculation that defines the phase.

Here we develop a reliability-aware active-learning workflow in which an audited exact BdG oracle remains the sole authority for thermodynamic and topological labels. We first reconstruct the two-dimensional Normal, uniform-superconducting, and FFLO structure while retaining a separate hard-risk numerical layer. On trusted gapped FFLO states, full-zone gap and Pfaffian diagnostics resolve cFFLO/tFFLO structure with an explicit neighborhood-size sensitivity audit. Finally, we extend the same oracle–scheduler separation to a three-dimensional parameter domain. The extension reveals thermodynamic and topology-aware structure across a finite sampled domain; its exact support is reported discretely and is not promoted to a continuous three-dimensional topology surface or a complete atlas.

Exact oracle and physical labels.—

For external coordinates $x = (k_BT/t, J_A/t, \mu/t, \ldots)$,

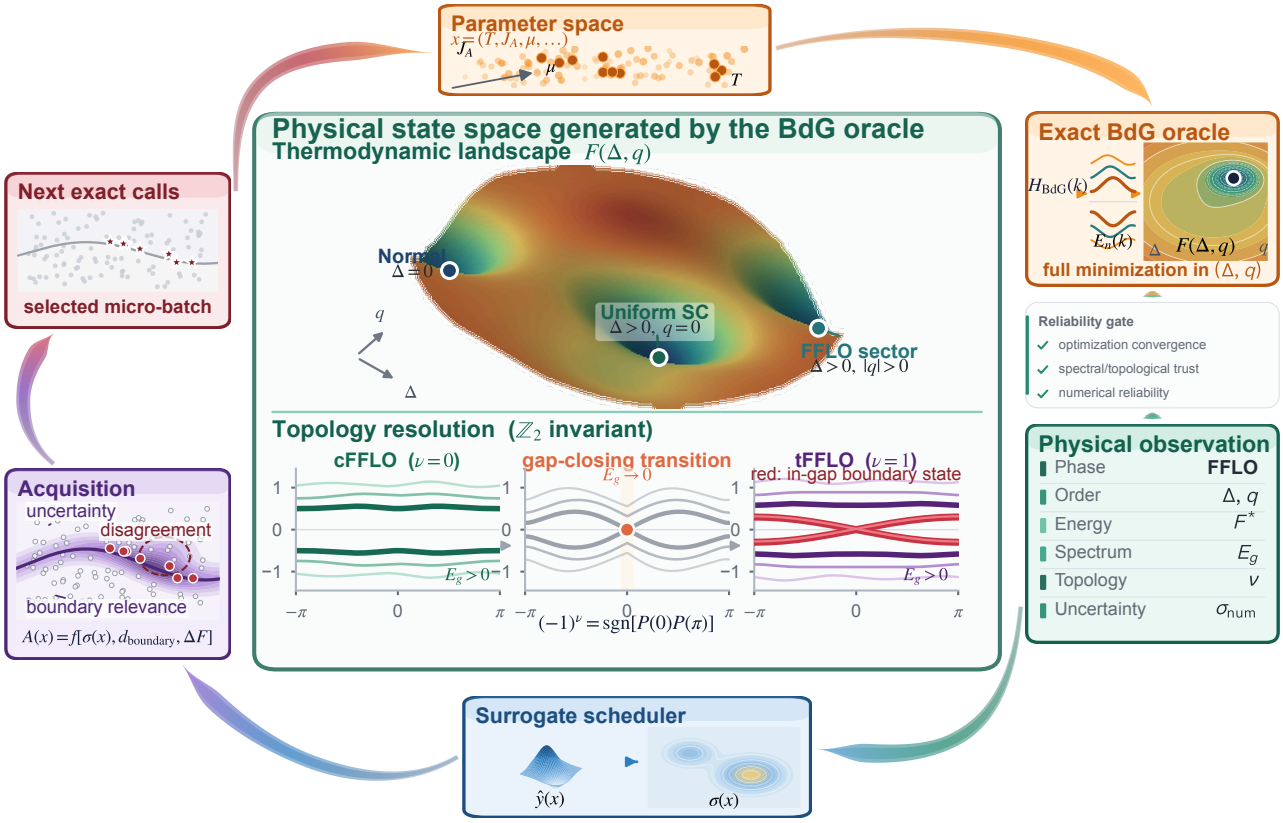


FIG. 1. **Reliability-aware exact-oracle and active-learning hierarchy.** The central schematic organizes the physical state space generated by the BdG oracle: representative minima of $F(\Delta, q)$ encode Normal ($\Delta = 0$), uniform-SC ($\Delta > 0, q = 0$), and FFLO ($\Delta > 0, |q| > 0$) sectors. Below, trusted gapped FFLO spectra distinguish cFFLO ($\nu = 0$), a bulk-gap-closing transition, and reopened gapped tFFLO ($\nu = 1$); the red crossing branch is an in-gap boundary state rather than a gapless bulk band. The right-hand oracle and reliability cards separate minimization, spectral/topological trust, and numerical status from the surrounding scheduler–acquisition cycle, which selects the next exact calls. All elements are schematic; the surrogate schedules BdG evaluations and does not determine physical labels or constitute convergence evidence.

the oracle uses the Nambu basis $\Psi_k = (c_{k+q/2,\uparrow}, c_{k+q/2,\downarrow}, c^\dagger_{-k+q/2,\uparrow}, c^\dagger_{-k+q/2,\downarrow})^T$ and the block Hamiltonian

$$H_{\mathrm{BdG}}(k) = \begin{pmatrix} h(k+q/2) - \mu & \Delta_q \\ \Delta_q^* & -\sigma_y h^T(-k+q/2)\sigma_y + \mu \end{pmatrix},$$
$$h(k) = t\cos k + (0, \lambda_y \sin k, \lambda_z \sin k + J_A \cos k) \cdot \boldsymbol{\sigma}. \tag{1}$$

At every exact call it constructs $F(\Delta, q)$, refines competing minima, and accepts the lowest reliable state only after comparison with the normal reference. The frozen thresholds $\Delta_\epsilon = 10^{-3}$ and $q_\epsilon = 10^{-2}$ separate Normal, uniform-SC, and FFLO observations; phase labels remain distinct from trusted, boundary-ambiguous, and rerun-required numerical states. Edge minima trigger adaptive q-window expansion and shallow Normal–SC competition triggers local Δ refinement, while unresolved outcomes are withheld from clean physical training labels. Ensemble surrogates rank the next exact micro-batch from accepted observations; they are schedulers rather than physics solvers (Fig. 1). Figure 1 is a schematic hierarchy and is not itself evidence that any convergence gate has closed.

Audited thermodynamic reconstruction.—

The two-dimensional loop begins from 512 selected exact calls, with 506 labels retained in the frozen endpoint, and subsequently allocates calls around visible Normal/superconducting and uniform-SC/FFLO structure (Fig. 2). The acquisition combines phase uncertainty, boundary proximity, phase-facing regression uncertainty, numerical risk, and soft repulsion, but only returned BdG calculations enter the accepted dataset. The endpoint contains 7,434 exact observations: 1,867 Normal, 715 uniform-SC, and 4,852 FFLO labels. Its final phase-map change is 0.001629, the Normal–SC and uniform-SC–FFLO shifts are 0.004167 and 0, and trusted surprise is 0 for 127 trusted calls, satisfying the configured stopping rules. The separate 129-call hard-risk layer, including 19 coordinate matches retained in the trusted endpoint, is a

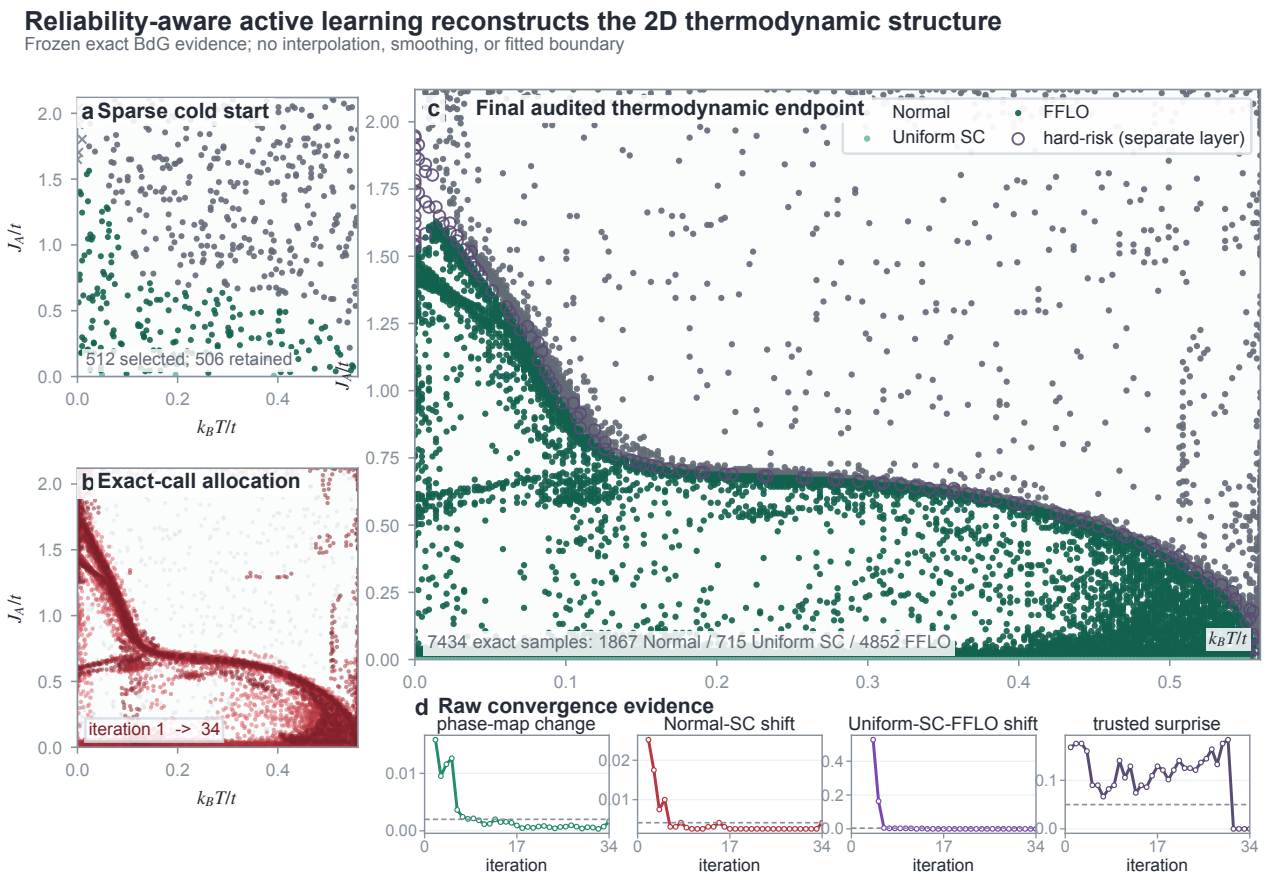


FIG. 2. **Audited two-dimensional thermodynamic reconstruction.** **(a)** Sparse random-grid cold start with 512 selected exact calls; 506 retained frozen labels are shown. **(b)** Iteration-resolved exact-call allocation over $(k_BT/t, J_A/t)$ concentrates around visible thermodynamic structure. **(c)** Exact-sample endpoint with 7,434 observations: 1,867 Normal, 715 uniform SC, and 4,852 FFLO. Open rings denote the separate 129-call hard-risk numerical layer, not a phase. **(d)** Raw stopping observables at iteration 34: phase-map change 0.001629 (threshold 0.002), Normal–SC shift 0.004167 (threshold 0.004167), uniform-SC–FFLO shift 0, and trusted surprise 0 for 127 trusted calls (threshold 0.05). No interpolated field or fitted boundary is used, and finite numerical risk remains explicit.

numerical frontier rather than a fourth phase or evidence that uncertainty has been eliminated. The exact-sample map therefore supports an audited two-dimensional thermodynamic reconstruction under the registered criterion while retaining finite numerical risk (Fig. 2).

Topology-resolved FFLO structure.—

Topology is evaluated only after thermodynamic acceptance and only on trusted gapped FFLO observations. A full-Brillouin-zone minimum-gap guard and Pfaffians P_0 and P_π determine the $\mathbb{Z}_2$ index away from their tolerances, assigning cFFLO for $\nu = 0$ and tFFLO for $\nu = 1$. Within the 4,345-point endpoint, 3,213 trusted gapped FFLO states divide into 2,290 cFFLO and 923 tFFLO observations and form a connected sampled structure (Fig. 3). The support-restricted contour and local opposite-$\mathbb{Z}_2$ brackets diagnose a gap-closing transition; neither is an exact boundary, and the closing is not a stable third phase. The configured $k = 8$ audit returns Decision A, with coverage $p_{95} = 0.006185 < 0.00625$ and the remaining registered metrics passing. For the $k = 6$ sensitivity setting, coverage $p_{95} = 0.006269$ lies slightly above the same threshold, so the configured pass retains a neighborhood-size sensitivity and does not exhaust all refinements or nodal alternatives (Fig. 3).

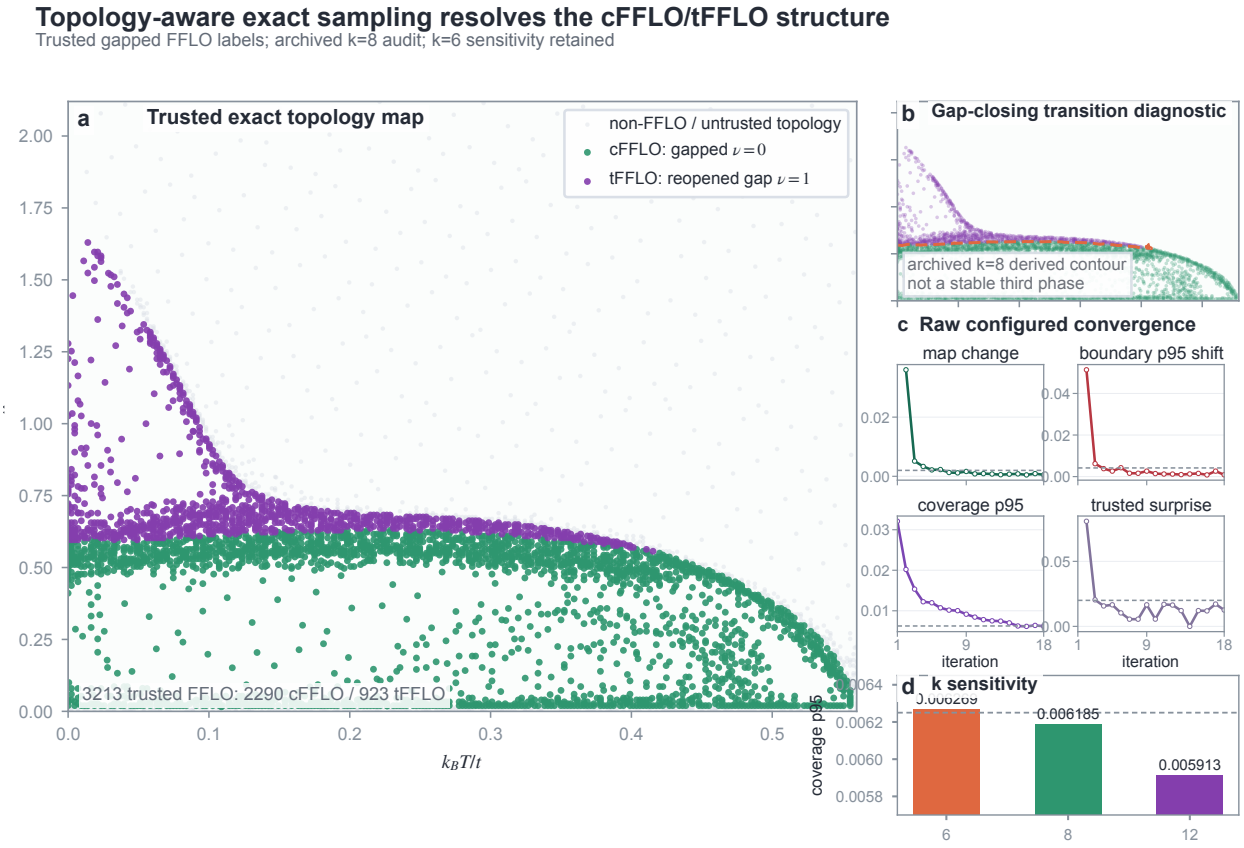


FIG. 3. **Topology-resolved cFFLO/tFFLO discovery and configured audit.** **(a)** Trusted gapped FFLO subset of the 4,345-point exact endpoint: 3,213 states comprising 2,290 cFFLO ($\nu = 0$) and 923 tFFLO ($\nu = 1$) labels. **(b)** The archived support-restricted $k = 8$ contour marks a gap-closing transition diagnostic; local opposite-$\mathbb{Z}_2$ brackets and the contour are supporting guides, not a stable phase or exact boundary. **(c)** Raw configured convergence histories for map change, boundary p_{95} shift, coverage p_{95}, and trusted surprise. **(d)** The configured $k = 8$ result gives Decision A with coverage $p_{95} = 0.006185 < 0.00625$, whereas the $k = 6$ sensitivity value 0.006269 lies slightly above the same threshold. Topology is assigned from full-Brillouin-zone gap/Pfaffian evidence; the audit does not establish that every possible refinement or nodal alternative is exhausted.

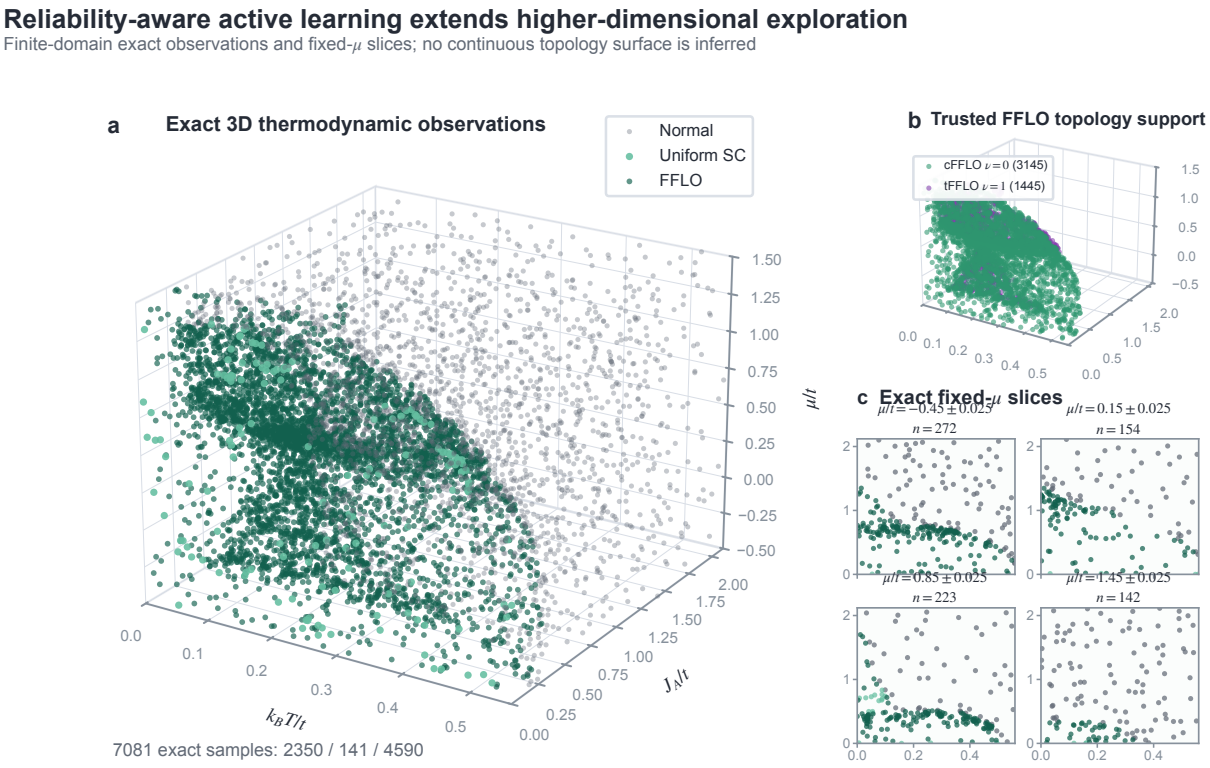


FIG. 4. **Active-learning-enabled higher-dimensional exploration. (a)** Exact three-dimensional thermodynamic point cloud in $(k_BT/t, J_A/t, \mu/t)$ over $\mu/t \in [-0.5, 1.5]$: 7,081 observations containing 2,350 Normal, 141 uniform-SC, and 4,590 FFLO states. **(b)** Trusted FFLO-only topology support comprises 3,145 cFFLO and 1,445 tFFLO points. **(c)** Four narrow fixed-μ bins, centered at $\mu/t = -0.45, 0.15, 0.85$, and 1.45 with half-width 0.025, show exact slice samples without interpolation between slices. The finite-domain support is discrete; no continuous topology surface or complete three-dimensional atlas is inferred.

Higher-dimensional exploration.—
The same oracle–scheduler separation extends the sampled coordinates to $x = (k_B T/t, J_A/t, \mu/t)$ over the finite interval $\mu/t \in [-0.5, 1.5]$. The endpoint contains 7,081 exact observations: 2,350 Normal, 141 uniform-SC, and 4,590 FFLO states (Fig. 4). Trusted FFLO-only topology support comprises 3,145 cFFLO and 1,445 tFFLO points and remains discrete rather than a fitted surface. Four explicit fixed-μ bins centered at $\mu/t = -0.45, 0.15, 0.85$, and 1.45, each with half-width 0.025, expose exact slice-to-slice structure without interpolation between slices. These point clouds and slices demonstrate higher-dimensional exploration and reveal thermodynamic and topology-aware structures across the sampled domain. Archived iteration histories are reported in the Supplemental Material as numerical stability diagnostics; no operational cutoff is promoted to a physical convergence criterion. Consequently, the main claim is bounded by the finite domain and discrete exact support, and does not imply a continuous topology surface or complete three-dimensional atlas (Fig. 4).

DISCUSSION

The combined results establish a reliability-aware active-learning workflow in which numerical trust remains part of each physical observation and exact BdG calculations retain label authority. The two-dimensional thermodynamic and configured topology audits establish the low-dimensional results, while the three-dimensional extension demonstrates why selective exact sampling becomes more consequential as the domain grows. That extension is deliberately bounded by its finite sampled interval, discrete exact support, and four fixed-μ slices; these data reveal structure but do not define a continuous topology surface or complete atlas.

The matched Stage-V audit likewise gives no overall acquisition winner: V3 has slightly denser topology-surface support ($p_{95} = 0.01893$ versus 0.02103), whereas V4 has lower sequential topology surprise (0.0925 versus 0.1189). V4 does not lower the common offline graph-uncertainty functional (0.01217 versus 0.01114), and none of its four candidate nodes has validated local same-identity support. Learned acquisition is therefore an operational exploratory mechanism, not evidence of superiority or autonomous new-phase discovery.

As a future direction, the Stage-VI studies indicate that thermodynamic state and $|q|$ are learnable and that continuous Pfaffian quantities become learnable after conditioning on trusted gapped superconductivity. The frozen export, however, lacks fixed-k spectra, a density-of-states proxy, and validated low-energy spectral moments, and its replay uncertainty is not uniformly monotonic. A ground-up physical world model must therefore validate those descriptors and calibrated uncertainty before production expected-information-gain acquisition can be authorized.

[1] P. Fulde and R. A. Ferrell, Physical Review **135**, A550 (1964).
[2] A. I. Larkin and Y. N. Ovchinnikov, Soviet Physics JETP **20**, 762 (1965).
[3] R. Casalbuoni and G. Nardulli, Reviews of Modern Physics **76**, 263 (2004).
[4] Y. Matsuda and H. Shimahara, Journal of the Physical Society of Japan **76**, 051005 (2007).
[5] J. J. Kinnunen, J. E. Baarsma, J.-P. Martikainen, and P. Törmä, Reports on Progress in Physics **81**, 046401 (2018).
[6] H. A. Radovan, N. A. Fortune, T. P. Murphy, S. T. Hannahs, E. C. Palm, S. W. Tozer, and D. Hall, Nature **425**, 51 (2003).
[7] A. Bianchi, R. Movshovich, C. Capan, P. G. Pagliuso, and J. L. Sarrao, Physical Review Letters **91**, 187004 (2003).
[8] M. Kenzelmann, T. Strässle, C. Niedermayer, M. Sigrist, B. Padmanabhan, M. Zolliker, A. D. Bianchi, R. Movshovich, E. D. Bauer, J. L. Sarrao, and J. D. Thompson, Science **321**, 1652 (2008).
[9] R. Lortz, Y. Wang, A. Demuer, P. H. M. Böttger, B. Bergk, G. Zwicknagl, Y. Nakazawa, and J. Wosnitza, Physical Review Letters **99**, 187002 (2007).
[10] H. Mayaffre, S. Krämer, M. Horvatić, C. Berthier, K. Miyagawa, K. Kanoda, and V. F. Mitrović, Nature Physics **10**, 928 (2014).
[11] Y. an Liao, A. S. C. Rittner, T. Paprotta, W. Li, G. B. Partridge, R. G. Hulet, S. K. Baur, and E. J. Mueller, Nature **467**, 567 (2010).
[12] M. W. Zwierlein, A. Schirotzek, C. H. Schunck, and W. Ketterle, Science **311**, 492 (2006).
[13] G. B. Partridge, W. Li, R. I. Kamar, Y. an Liao, and R. G. Hulet, Science **311**, 503 (2006).
[14] D. E. Sheehy and L. Radzihovsky, Annals of Physics **322**, 1790 (2007).
[15] L. Radzihovsky and D. E. Sheehy, Reports on Progress in Physics **73**, 076501 (2010).
[16] L. Šmejkal, J. Sinova, and T. Jungwirth, Physical Review X **12**, 040501 (2022).
[17] L. Šmejkal, J. Sinova, and T. Jungwirth, Physical Review X **12**, 031042 (2022).
[18] I. Mazin, Physical Review X **12**, 040002 (2022).
[19] L. Šmejkal, R. González-Hernández, T. Jungwirth, and J. Sinova, Science Advances **6**, eaaz8809 (2020).
[20] D. Chakraborty and A. M. Black-Schaffer, Physical Review B **110**, L060508 (2024).
[21] S. Hong, M. J. Park, and K.-M. Kim, Physical Review B **111**, 054501 (2025).
[22] A. Bose, S. Vadnais, and A. Paramekanti, Physical Review B **110**, 205120 (2024).
[23] D. Zhu, Z.-Y. Zhuang, Z. Wu, and Z. Yan, Physical Review B **108**, 184505 (2023).
[24] S. A. A. Ghorashi, T. L. Hughes, and J. Cano, Physical Review Letters **133**, 106601 (2024).
[25] W. Zhang and W. Yi, Nature Communications **4**, 2711 (2013).
[26] C. Qu, Z. Zheng, M. Gong, Y. Xu, L. Mao, X. Zou,

G. Guo, and C. Zhang, Nature Communications **4**, 2710 (2013).
[27] Y. Cao, S.-H. Zou, X.-J. Liu, S. Yi, G.-L. Long, and H. Hu, Physical Review Letters **113**, 115302 (2014).
[28] N. Read and D. Green, Physical Review B **61**, 10267 (2000).
[29] M. Sato and Y. Ando, Reports on Progress in Physics **80**, 076501 (2017).
[30] X.-L. Qi and S.-C. Zhang, Reviews of Modern Physics **83**, 1057 (2011).
[31] A. P. Schnyder, S. Ryu, A. Furusaki, and A. W. W. Ludwig, Physical Review B **78**, 195125 (2008).
[32] S. Ryu, A. P. Schnyder, A. Furusaki, and A. W. W. Ludwig, New Journal of Physics **12**, 065010 (2010).
[33] R. M. Lutchyn, J. D. Sau, and S. D. Sarma, Physical Review Letters **105**, 077001 (2010).
[34] Y. Oreg, G. Refael, and F. von Oppen, Physical Review Letters **105**, 177002 (2010).
[35] J. Alicea, Reports on Progress in Physics **75**, 076501 (2012).
[36] C. Beenakker, Annual Review of Condensed Matter Physics **4**, 113 (2013).
[37] A. Y. Kitaev, Physics-Uspekhi **44**, 131 (2001).
[38] J. C. Budich and E. Ardonne, Physical Review B **88**, 075419 (2013).
[39] D. Cohn, L. Atlas, and R. Ladner, Machine Learning **15**, 201 (1994).
[40] D. J. C. MacKay, Neural Computation **4**, 590 (1992).
[41] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, and N. de Freitas, Proceedings of the IEEE **104**, 148 (2016).
[42] K. Terayama, R. Tamura, Y. Nose, H. Hiramatsu, H. Hosono, Y. Okuno, and K. Tsuda, Physical Review Materials **3**, 033802 (2019).
[43] C. Dai, I. R. Bruss, and S. C. Glotzer, The Journal of Physical Chemistry B **124**, 1275 (2020).
[44] T. Lookman, P. V. Balachandran, D. Xue, and R. Yuan, npj Computational Materials **5**, 21 (2019).
[45] K. Tran and Z. W. Ulissi, Nature Catalysis **1**, 696 (2018).
[46] E. V. Podryabinkin and A. V. Shapeev, Computational Materials Science **140**, 171 (2017).
[47] R. Jinnouchi, F. Karsai, and G. Kresse, Physical Review B **100**, 014105 (2019).
[48] J. Vandermause, S. B. Torrisi, S. Batzner, Y. Xie, L. Sun, A. M. Kolpak, and B. Kozinsky, npj Computational Materials **6**, 20 (2020).
[49] R. Batra, L. Song, and R. Ramprasad, Nature Reviews Materials **6**, 655 (2021).
[50] F. Noé, S. Olsson, J. Köhler, and H. Wu, Science **365**, eaaw1147 (2019).